

## #HW 6

● Graded

42 Days, 59 Minutes Late

Student

Diego Negrete

Total Points

75 / 100 pts

Question 1

1

50 / 50 pts

✓ - 0 pts correct

Question 2

2

25 / 50 pts

- 0 pts Correct

- 2 pts PV =  $mZRT/M$ , so you should also include the molecular weight in the calculation. The remaining procedure is correct.

- 20 pts incomplete

✓ - 25 pts late

No questions assigned to the following page.

Hws # 6

Question #1

**Case Study 1: Polytropic Compression in Green Ammonia Production****Background:**

Green ammonia is emerging as a key energy carrier in decarbonized economies. Compressors used in ammonia synthesis must handle gas efficiently — and property evaluation is essential for performance prediction.

**Problem:**

Nitrogen gas enters a polytropic compressor at 100 kPa and 300 K and is compressed to 2 MPa following  $Pv^n = \text{constant}$  with  $n = 1.25$ .

**Tasks:**

- Using the ideal gas law and polytropic relations, calculate the final temperature and specific work input.
- Use real gas data (e.g., compressibility factor) to improve your calculation.
- Solve the problem using an AI tool.
- Compare and discuss differences in your results and the AI's.

**Reflection:**

How would errors in property evaluation influence compressor design, cost, and sustainability in green ammonia production plants?

$$P_1 = 100 \text{ kPa} \quad \left\{ \begin{array}{l} \text{compressed} \\ T_1 = 300 \text{ K} \end{array} \right. \quad \left\{ \begin{array}{l} P_2 = 2 \text{ MPa} \\ T_2 = ? \end{array} \right.$$

Ideal Gas Law

$$Pv = RT \quad ; \quad R = 8.314 \frac{\text{kJ}}{\text{kmol} \cdot \text{K}}$$

$$100 \text{ kPa} \cdot v = (8.314 \frac{\text{kJ}}{\text{kmol} \cdot \text{K}}) (300 \text{ K}) \left( \frac{1}{28.01 \frac{\text{kg}}{\text{kmol}}} \right)$$

$$\Rightarrow v = \frac{(8.314 \frac{\text{kJ}}{\text{kmol} \cdot \text{K}}) (300 \text{ K}) \left( \frac{1}{28.01 \frac{\text{kg}}{\text{kmol}}} \right)}{100 \text{ kPa}} = \frac{89.097 \frac{\text{kJ}}{\text{kg}}}{100 \text{ kPa}} = \frac{89.097 \frac{\text{m}^3 \cdot \text{Pa}}{\text{kg}}}{100 \text{ kPa}} = 0.89097 \text{ m}^3/\text{kg}$$

$$\Rightarrow W = \int_{v_1}^{v_2} P dv \approx \frac{W}{m} = \int_{v_1}^{v_2} P dv$$

$$P \cdot v^n = \text{constant} \Rightarrow P = \frac{\text{constant}}{v^n}$$

$$\frac{W}{m} = \int_{v_1}^{v_2} \frac{\text{constant}}{v^n} dv \Rightarrow \int_{0.89097}^{0.0806} \frac{0.0806}{v^{1.25}} dv \Rightarrow 0.365 \int_{0.89097}^{0.0806} \frac{1}{v^{1.25}} dv \Rightarrow 0.365 \left[ -4v^{-0.25} \right]_{0.89097}^{0.0806}$$

$$\Rightarrow 0.365 \left[ -4(0.0806)^{-0.25} - (-4(0.89097)^{-0.25}) \right] = -2.923 \text{ bar} \cdot \text{m}^3/\text{kg}$$

$$-2.923 \frac{\text{bar} \cdot \text{m}^3}{\text{kg}} \cdot \frac{10^5 \text{ Pa}}{1 \text{ bar}} = -2.923 \times 10^5 \frac{\text{Pa} \cdot \text{m}^3}{\text{kg}} = -2.923 \times 10^5 \frac{\text{Joules}}{\text{kg}} \cdot \frac{1 \text{ kJ}}{1000 \text{ J}} = -292.3 \frac{\text{kJ}}{\text{kg}}$$

$$b) \quad Pv = RT$$

$$P_1 = 1 \text{ bar} \Rightarrow P_2 = 20 \text{ bar}$$

$$T_1 = 300 \text{ K} \Rightarrow T_2 = 546.23 \text{ K}$$

$$\frac{P_1}{P_2} = \frac{v_2}{v_1} = \frac{1 \text{ bar}}{20 \text{ bar}} = 0.05$$

$$\frac{T_1}{T_2} = \frac{P_1}{P_2} = \frac{300 \text{ K}}{166 \text{ K}} = 2.581$$

From compressibility chart,  $Z = 0.98$ From compressibility chart,  $Z = 0.88$ 

$$Pv = RT \Rightarrow v_1 = \frac{RT_1}{P_1} = \frac{(8.314 \frac{\text{kJ}}{\text{kmol} \cdot \text{K}}) (300 \text{ K})}{100 \text{ kPa}} = 0.891 \text{ m}^3/\text{kg}$$

$$\Rightarrow v_2 = \frac{RT_2}{P_2} = \frac{(8.314 \frac{\text{kJ}}{\text{kmol} \cdot \text{K}}) (546.23 \text{ K})}{2000 \text{ kPa}} = 0.0713 \text{ m}^3/\text{kg}$$

$$P_1 v_1^n = P_2 v_2^n$$

$$(100 \text{ kPa}) (0.891 \text{ m}^3/\text{kg})^{1.25} = 0.854$$

c) Using AI Tool (ChatGPT)

Ideal Gas Result:  $292 \text{ kJ/kg}$   
Real Gas Result:  $291-292 \text{ kJ/kg}$

d) Compare + Contrast Between AI and myself

→ For part a, AI and I came up with the exact same answer. However, for part b, this was a difference in answers. I came up with a higher value. This can be due to the  $Z_2$  values we came up with. From the  $Z$ -charts chart, the  $Z_1$  and  $Z_2$  values indicate the value of  $Z_1$  is 0.88. ChatGPT came up with 0.98 from another chart source.

$$\text{Polytropic Process} \quad \left\{ \begin{array}{l} P v^n = \text{constant} ; n = 1.25 \\ P \cdot v^n = \text{constant} ; n = 1.25 \end{array} \right.$$

For State 1:  $P_1 v_1^n = \text{constant}$ 

$$\Rightarrow 100 \text{ kPa} \cdot \left( \frac{0.89097 \text{ m}^3/\text{kg}}{100 \text{ kPa}} \right)^{1.25} = \text{constant}$$

$$\Rightarrow (1 \text{ bar}) (0.89097 \text{ m}^3/\text{kg})^{1.25} = \text{constant}$$

$$\Rightarrow \text{constant} = 0.854$$

use this value to calc. the unknown  $v_2$  using  $P \cdot v^n = \text{constant}$

For State 2:  $P_2 v_2^n = \text{constant}$ 

$$\Rightarrow 2 \text{ MPa} \cdot \left( \frac{v_2}{100 \text{ kPa}} \right)^{1.25} = 20 \text{ bar}$$

$$\Rightarrow (20 \text{ bar}) (v_2)^{1.25} = 0.854$$

$$\Rightarrow v_2^{1.25} = 0.04825$$

$$\Rightarrow (v_2)^{1.25} = 0.04825 \text{ m}^3/\text{kg}$$

$$\Rightarrow v_2 = 0.0806 \text{ m}^3/\text{kg}$$

use  $v_1$  +  $v_2$  to calc. work

$$Pv = RT \quad ; \quad P_2 = 2 \text{ MPa} \quad \frac{1 \text{ m}^3 \cdot \text{Pa}}{1 \text{ bar}} \cdot \frac{10^5 \text{ Pa}}{1 \text{ bar}} = 2000 \text{ kPa}$$

$$\Rightarrow P_2 v_2 = R T_2$$

$$\Rightarrow 2000 \text{ kPa} \cdot (0.0806 \text{ m}^3/\text{kg}) = (8.314 \frac{\text{kJ}}{\text{kmol} \cdot \text{K}}) (T_2) \left( \frac{1}{28.01 \frac{\text{kg}}{\text{kmol}}} \right)$$

$$\Rightarrow 2000 \text{ kPa} \cdot (0.0806 \text{ m}^3/\text{kg}) = 0.2968 \frac{\text{kJ}}{\text{kg}} (T_2)$$

$$\Rightarrow T_2 = \frac{2000 \text{ kPa} \cdot (0.0806 \text{ m}^3/\text{kg})}{0.2968 \frac{\text{kJ}}{\text{kg}}} = 546.23 \text{ K}$$

**Reflection:**

→ Compressor design influence

- Parameters such as polytropic work heavily rely on accurate property values. If these values are inaccurate, this can affect the compressors operational stability and performance.

→ Cost influence

- Property values such as temperature and pressure can easily be evaluated inaccurately which can lead to the "need" for more expensive equipment, ultimately leading to an increase of materials.

→ Sustainability influence

- Incorrect designs caused by inaccurate property values can lead to an unnecessary use of materials causing the waste of materials and the energy it took to run the design process.

No questions assigned to the following page.

## Question #2

### Background

In hydrogen storage and distribution systems, property evaluation accuracy is critical for safety. Miscalculating gas volume or storage capacity – for example, by assuming ideal gas behavior when real gas effects are significant – can lead to tank overpressure, pressure relief valve (PRV) activation, or even leaks and explosions. In 2019, a hydrogen fueling station in Kjørbø, Norway, exploded after a hydrogen release ignited, underscoring how quickly hydrogen can form a flammable cloud and ignite. One frequent cause of incidents is incorrectly calculating how much gas a tank can safely receive. This case study explores a simplified version of such a scenario. You will calculate hydrogen storage limits using both the ideal gas law and real-gas compressibility, then analyze how these differences could lead to unsafe conditions.

### Problem

- A fixed-roof hydrogen buffer tank upstream of a compressor is designed as follows:
- Internal volume:  $V_{\text{tank}} = 50.0 \text{ m}^3$
  - Maximum allowable working pressure (MAWP): 200 bar (gauge)
  - PSV (pressure safety valve) set pressure: 200 bar (gauge)

- Ambient design temperature:  $25^\circ\text{C}$
- Isotoping gas temperature: up to  $40^\circ\text{C}$
- Initially tank has dry hydrogen at 25 bar (g) and at  $25^\circ\text{C}$ .

### Tasks

- Calculate the current mass of hydrogen in the tank:
  - (a) using the ideal gas law
  - (b) using the real-gas equation with an appropriate Z value. (Find Z values from textbook's chart and from this website <https://schools.org/physics/hydrogen-data/hydrogen-compressibility-different-temperatures-and-pressures> (data source: chugat.com))
- Assuming ideal behavior and a final tank pressure of 200 bar (g) at  $25^\circ\text{C}$ , calculate the maximum additional mass of hydrogen the operator believes the tank can receive.
- Repeat steps above using a compressibility factor Z appropriate for 200 bar. Compare the result to the ideal-gas calculation and find the percentage error.
- Suppose the gas inside the tank later warms to  $40^\circ\text{C}$  after filling.
  - (a) if the operator's calculated mass is added, what pressure will the tank reach (ideal-gas assumption)?
  - (b) What pressure will it reach with real-gas behavior?
  - (c) Solve for the safe mass of hydrogen that ensures the final pressure does not exceed 200 bar (g) at  $40^\circ\text{C}$  using real-gas assumptions.
- If the tank is overfilled and the PSV vents for 2 minutes at an average flow rate of  $0.5 \text{ kg/s}$ , estimate the total mass released. Briefly discuss why a hydrogen release of this magnitude is hazardous.
- Use ChatGPT (or another large language model) to solve the same problem. Your submission must include:
  - The exact prompt you used
  - The complete output from the AI tool (included as an appendix)
  - Ask the AI to:
    - Use both ideal-gas and real-gas equations.
    - State all assumptions and values of Z.
    - Solve the problem at both  $25^\circ\text{C}$  and  $40^\circ\text{C}$ .

### Reflection and Discussion

- Answer the following in 1–2 paragraphs:
- How do your hand-calculated results differ from the AI's? What caused the differences?
  - When did the ideal-gas assumption lead to significant error, and why?
  - What engineering or safety systems could prevent an incident caused by overfilling or overpressure (e.g., instrumentation, procedures, design features)?
  - What does this case teach about the importance of property evaluation and engineering judgment in real-world design?

1a)  $P_{\text{abs}} = P_{\text{gauge}} + P_{\text{atm}} = 25 \text{ bar} + 1 \text{ bar} = 26 \text{ bar}$

$PV = nRT$

$PV = nRT; T = 25^\circ\text{C} = 298.15 \text{ K}$

$$(26 \text{ bar}) (50 \text{ m}^3) = n (8.314 \frac{\text{J}}{\text{mol} \cdot \text{K}}) (298.15 \text{ K})$$

$$\Rightarrow n = \frac{(26 \text{ bar}) (50 \text{ m}^3)}{(8.314 \frac{\text{J}}{\text{mol} \cdot \text{K}}) (298.15 \text{ K})} = 105.781 \text{ kg}$$

1b) Real Gas Law

$PV = nZRT$

$PV = nZRT \Rightarrow n = \frac{PV}{ZRT}; Z = \frac{P}{P_a} = \frac{26 \text{ bar}}{1 \text{ bar}} = 26$

$Z = \frac{P}{P_a} = \frac{26 \text{ bar}}{1 \text{ bar}} = 26$

Interpolation from Hydrogen Table Data Chart

$P = 26 \text{ MPa}$

$T = 25^\circ\text{C}$

$Z = 1.0543$

$n = \frac{PV}{ZRT} = \frac{(26 \text{ bar}) (50 \text{ m}^3)}{(1.0543) (8.314 \frac{\text{J}}{\text{mol} \cdot \text{K}}) (298.15 \text{ K})} = 109.77 \text{ kg}$

2. Assuming Ideal Behavior:  $P = 200 \text{ bar}, T = 25^\circ\text{C}$

$P_{\text{abs}} = P_{\text{gauge}} + P_{\text{atm}} = (200 \text{ bar}) + 1 \text{ bar} = 201 \text{ bar}$

$PV = nRT$

$n = \frac{PV}{RT} = \frac{(201 \text{ bar}) (50 \text{ m}^3)}{(8.314 \frac{\text{J}}{\text{mol} \cdot \text{K}}) (298.15 \text{ K})} = 1095000 \text{ mol}$

$m = nM = (1095000 \text{ mol}) (2.016 \frac{\text{g}}{\text{mol}}) = 2208 \text{ kg}$

Maximum Additional Mass:  $m_{\text{add}} = m_{\text{total}} - m_{\text{initial}} = 2208 \text{ kg} - 105.781 \text{ kg} = 2102.219 \text{ kg}$

3. Assuming Real Gas Behavior:  $P = 200 \text{ bar}, T = 25^\circ\text{C}$

$PV = nZRT$

$n = \frac{PV}{ZRT} = \frac{(201 \text{ bar}) (50 \text{ m}^3)}{(1.0543) (8.314 \frac{\text{J}}{\text{mol} \cdot \text{K}}) (298.15 \text{ K})} = 221.191 \text{ kg}$

$m = nM = (221.191 \text{ kg}) (2.016 \frac{\text{g}}{\text{mol}}) = 445.8 \text{ kg}$

Maximum Additional Mass:  $m_{\text{add}} = m_{\text{total}} - m_{\text{initial}} = (221.191 - 105.781) \text{ kg} = 115.41 \text{ kg}$

### Hydrogen Analysis Resource Center: Hydrogen Compressibility (Z) at Different Temperatures ( $^\circ\text{C}$ ) and Pressures (MPa)

| Temperature ( $^\circ\text{C}$ ) | 0.1    | 1      | 10     | 20     | 30     | 40     | 50     | 60     | 70     | 80     | 90     | 100    |
|----------------------------------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| -150                             | 1.0003 | 1.0008 | 1.0019 | 1.0028 | 1.0037 | 1.0047 | 1.0057 | 1.0067 | 1.0077 | 1.0087 | 1.0097 | 1.0107 |
| -100                             | 1.0007 | 1.0016 | 1.0036 | 1.0056 | 1.0076 | 1.0096 | 1.0116 | 1.0136 | 1.0156 | 1.0176 | 1.0196 | 1.0216 |
| -50                              | 1.0017 | 1.0036 | 1.0066 | 1.0106 | 1.0146 | 1.0186 | 1.0226 | 1.0266 | 1.0306 | 1.0346 | 1.0386 | 1.0426 |
| 0                                | 1.0037 | 1.0077 | 1.0137 | 1.0217 | 1.0307 | 1.0407 | 1.0507 | 1.0607 | 1.0707 | 1.0807 | 1.0907 | 1.1007 |
| 50                               | 1.0086 | 1.0146 | 1.0226 | 1.0326 | 1.0446 | 1.0576 | 1.0716 | 1.0856 | 1.0996 | 1.1136 | 1.1276 | 1.1416 |
| 100                              | 1.0166 | 1.0246 | 1.0346 | 1.0466 | 1.0606 | 1.0756 | 1.0916 | 1.1076 | 1.1236 | 1.1396 | 1.1556 | 1.1716 |
| 150                              | 1.0276 | 1.0376 | 1.0496 | 1.0636 | 1.0796 | 1.0966 | 1.1136 | 1.1306 | 1.1476 | 1.1646 | 1.1816 | 1.1986 |
| 200                              | 1.0416 | 1.0536 | 1.0676 | 1.0836 | 1.1016 | 1.1206 | 1.1396 | 1.1586 | 1.1776 | 1.1966 | 1.2156 | 1.2346 |
| 250                              | 1.0576 | 1.0706 | 1.0856 | 1.1026 | 1.1216 | 1.1416 | 1.1616 | 1.1816 | 1.2016 | 1.2216 | 1.2416 | 1.2616 |
| 300                              | 1.0756 | 1.0896 | 1.1056 | 1.1236 | 1.1436 | 1.1646 | 1.1856 | 1.2066 | 1.2266 | 1.2466 | 1.2666 | 1.2866 |
| 350                              | 1.0956 | 1.1106 | 1.1276 | 1.1466 | 1.1676 | 1.1886 | 1.2096 | 1.2306 | 1.2506 | 1.2706 | 1.2906 | 1.3106 |
| 400                              | 1.1176 | 1.1336 | 1.1516 | 1.1716 | 1.1926 | 1.2146 | 1.2366 | 1.2586 | 1.2806 | 1.3026 | 1.3246 | 1.3466 |
| 450                              | 1.1416 | 1.1586 | 1.1776 | 1.1986 | 1.2206 | 1.2436 | 1.2666 | 1.2896 | 1.3126 | 1.3356 | 1.3586 | 1.3816 |
| 500                              | 1.1676 | 1.1856 | 1.2056 | 1.2276 | 1.2506 | 1.2746 | 1.2986 | 1.3226 | 1.3466 | 1.3706 | 1.3946 | 1.4186 |
| 550                              | 1.1956 | 1.2146 | 1.2356 | 1.2586 | 1.2826 | 1.3076 | 1.3326 | 1.3576 | 1.3826 | 1.4076 | 1.4326 | 1.4576 |
| 600                              | 1.2256 | 1.2456 | 1.2676 | 1.2916 | 1.3166 | 1.3426 | 1.3686 | 1.3946 | 1.4206 | 1.4466 | 1.4726 | 1.4986 |
| 650                              | 1.2576 | 1.2786 | 1.3016 | 1.3266 | 1.3526 | 1.3786 | 1.4046 | 1.4306 | 1.4566 | 1.4826 | 1.5086 | 1.5346 |
| 700                              | 1.2916 | 1.3136 | 1.3376 | 1.3636 | 1.3906 | 1.4176 | 1.4446 | 1.4716 | 1.4986 | 1.5256 | 1.5526 | 1.5796 |
| 750                              | 1.3276 | 1.3506 | 1.3756 | 1.4026 | 1.4306 | 1.4586 | 1.4866 | 1.5146 | 1.5426 | 1.5706 | 1.5986 | 1.6266 |
| 800                              | 1.3656 | 1.3896 | 1.4156 | 1.4436 | 1.4726 | 1.5016 | 1.5306 | 1.5596 | 1.5886 | 1.6176 | 1.6466 | 1.6756 |
| 850                              | 1.4056 | 1.4306 | 1.4576 | 1.4856 | 1.5146 | 1.5436 | 1.5726 | 1.6016 | 1.6306 | 1.6596 | 1.6886 | 1.7176 |
| 900                              | 1.4476 | 1.4736 | 1.5016 | 1.5306 | 1.5606 | 1.5906 | 1.6206 | 1.6506 | 1.6806 | 1.7106 | 1.7406 | 1.7706 |
| 950                              | 1.4916 | 1.5186 | 1.5476 | 1.5776 | 1.6086 | 1.6396 | 1.6706 | 1.7016 | 1.7326 | 1.7636 | 1.7946 | 1.8256 |
| 1000                             | 1.5376 | 1.5656 | 1.5956 | 1.6266 | 1.6576 | 1.6886 | 1.7196 | 1.7506 | 1.7816 | 1.8126 | 1.8436 | 1.8746 |

5. Ideal Gas Assumption

$T = 40^\circ\text{C} = 313.15 \text{ K}$

$m_{\text{add}} = 819.77 \text{ kg}$

$PV = nRT$

$n = \frac{PV}{RT} = \frac{(201 \text{ bar}) (50 \text{ m}^3)}{(8.314 \frac{\text{J}}{\text{mol} \cdot \text{K}}) (313.15 \text{ K})} = 3849.1 \text{ mol}$

$m_{\text{add}} = nM = (3849.1 \text{ mol}) (2.016 \frac{\text{g}}{\text{mol}}) = 7760.8 \text{ g} = 7.7608 \text{ kg}$

$P_{\text{abs}} = P_{\text{gauge}} + P_{\text{atm}} = (200 \text{ bar}) + 1 \text{ bar} = 201 \text{ bar}$

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$n = \frac{PV}{RT} = \frac{(201 \text{ bar}) (50 \text{ m}^3)}{(8.314 \frac{\text{J}}{\text{mol} \cdot \text{K}}) (313.15 \text{ K})} = 3849.1 \text{ mol}$

$m_{\text{add}} = nM = (3849.1 \text{ mol}) (2.016 \frac{\text{g}}{\text{mol}}) = 7760.8 \text{ g} = 7.7608 \text{ kg}$

$P_{\text{abs}} = P_{\text{gauge}} + P_{\text{atm}} = (200 \text{ bar}) + 1 \text{ bar} = 201 \text{ bar}$

$PV = nRT$

$n = \frac{PV}{RT} = \frac{(201 \text{ bar}) (50 \text{ m}^3)}{(8.314 \frac{\text{J}}{\text{mol} \cdot \text{K}}) (313.15 \text{ K})} = 3849.1 \text{ mol}$

$m_{\text{add}} = nM = (3849.1 \text{ mol}) (2.016 \frac{\text{g}}{\text{mol}}) = 7760.8 \text{ g} = 7.7608 \text{ kg}$

Page 1

USE OF REAL GAS EQUATION

Page 1 of 1

# Use the Real Gas Equation for Real Gases

The Real Gas Equation of state for a real gas is:

$$PV = nZRT$$

where Z is the compressibility factor, which is a function of temperature and pressure.

Compressibility Factor (Z) values for various gases at different temperatures and pressures:

| Gas      | Temperature ( $^\circ\text{C}$ ) | Pressure (MPa) | Z      |
|----------|----------------------------------|----------------|--------|
| Hydrogen | -150                             | 0.1            | 1.0003 |
| Hydrogen | -100                             |                |        |
| Hydrogen | -50                              |                |        |
| Hydrogen | 0                                |                |        |
| Hydrogen | 50                               |                |        |
| Hydrogen | 100                              |                |        |
| Hydrogen | 150                              |                |        |
| Hydrogen | 200                              |                |        |
| Hydrogen | 250                              |                |        |
| Hydrogen | 300                              |                |        |
| Hydrogen | 350                              |                |        |
| Hydrogen | 400                              |                |        |
| Hydrogen | 450                              |                |        |
| Hydrogen | 500                              |                |        |
| Hydrogen | 550                              |                |        |
| Hydrogen | 600                              |                |        |
| Hydrogen | 650                              |                |        |
| Hydrogen | 700                              |                |        |
| Hydrogen | 750                              |                |        |
| Hydrogen | 800                              |                |        |
| Hydrogen | 850                              |                |        |
| Hydrogen | 900                              |                |        |
| Hydrogen | 950                              |                |        |
| Hydrogen | 1000                             |                |        |
| Hydrogen | 1050                             |                |        |
| Hydrogen | 1100                             |                |        |
| Hydrogen | 1150                             |                |        |
| Hydrogen | 1200                             |                |        |
| Hydrogen | 1250                             |                |        |
| Hydrogen | 1300                             |                |        |
| Hydrogen | 1350                             |                |        |
| Hydrogen | 1400                             |                |        |
| Hydrogen | 1450                             |                |        |
| Hydrogen | 1500                             |                |        |
| Hydrogen | 1550                             |                |        |
| Hydrogen | 1600                             |                |        |
| Hydrogen | 1650                             |                |        |
| Hydrogen | 1700                             |                |        |
| Hydrogen | 1750                             |                |        |
| Hydrogen | 1800                             |                |        |
| Hydrogen | 1850                             |                |        |
| Hydrogen | 1900                             |                |        |
| Hydrogen | 1950                             |                |        |
| Hydrogen | 2000                             |                |        |
| Hydrogen | 2050                             |                |        |
| Hydrogen | 2100                             |                |        |
| Hydrogen | 2150                             |                |        |
| Hydrogen | 2200                             |                |        |
| Hydrogen | 2250                             |                |        |
| Hydrogen | 2300                             |                |        |
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| Hydrogen | 3000                             |                |        |
| Hydrogen | 3050                             |                |        |
| Hydrogen | 3100                             |                |        |
| Hydrogen | 3150                             |                |        |
| Hydrogen | 3200                             |                |        |
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| Hydrogen | 3950                             |                |        |
| Hydrogen | 4000                             |                |        |
| Hydrogen | 4050                             |                |        |
| Hydrogen | 4100                             |                |        |
| Hydrogen | 4150                             |                |        |
| Hydrogen | 4200                             |                |        |
| Hydrogen | 4250                             |                |        |
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| Hydrogen | 4350                             |                |        |
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| Hydrogen | 6900                             |                |        |
| Hydrogen | 6950                             |                |        |
| Hydrogen | 7000                             |                |        |
| Hydrogen | 7050                             |                |        |
| Hydrogen | 7100                             |                |        |
| Hydrogen | 7150                             |                |        |
| Hydrogen | 7200                             |                |        |
| Hydrogen | 7250                             |                |        |
| Hydrogen | 7300                             |                |        |
| Hydrogen | 7350                             |                |        |
| Hydrogen | 7400                             |                |        |
| Hydrogen | 7450                             |                |        |
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| Hydrogen | 7900                             |                |        |
| Hydrogen | 7950                             |                |        |
| Hydrogen | 8000                             |                |        |
| Hydrogen | 8050                             |                |        |
| Hydrogen | 8100                             |                |        |
| Hydrogen | 8150                             |                |        |
| Hydrogen | 8200                             |                |        |
| Hydrogen | 8250                             |                |        |
| Hydrogen | 8300                             |                |        |
| Hydrogen | 8350                             |                |        |
| Hydrogen | 8400                             |                |        |
| Hydrogen | 8450                             |                |        |
| Hydrogen | 8500                             |                |        |
| Hydrogen | 8550                             |                |        |
| Hydrogen | 8600                             |                |        |
| Hydrogen | 8650                             |                |        |
| Hydrogen | 8700                             |                |        |
| Hydrogen | 8750                             |                |        |
| Hydrogen | 8800                             |                |        |
| Hydrogen | 8850                             |                |        |
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| Hydrogen | 8950                             |                |        |
| Hydrogen | 9000                             |                |        |
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| Hydrogen | 9150                             |                |        |
| Hydrogen | 9200                             |                |        |
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| Hydrogen | 9350                             |                |        |
| Hydrogen | 9400                             |                |        |
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| Hydrogen | 9800                             |                |        |
| Hydrogen | 9850                             |                |        |
| Hydrogen | 9900                             |                |        |
| Hydrogen | 9950                             |                |        |
| Hydrogen | 10000                            |                |        |

(10)

No questions assigned to the following page.

### Background

Development of a predictive model of distribution systems, property characteristics and occupancy is critical for the successful design of a building. The model must be able to predict the energy loads of the building and the energy consumption of the building. The model must be able to predict the energy loads of the building and the energy consumption of the building. The model must be able to predict the energy loads of the building and the energy consumption of the building.

### Results

A model of building before test processes of a computer is designed as follows:

- The model is a model of building before test processes of a computer.
- The model is a model of building before test processes of a computer.
- The model is a model of building before test processes of a computer.

Authoritative design temperature: 25 °C

Authoritative design humidity ratio: 10 g/kg

Authoritative design wind speed: 10 m/s

Authoritative design solar radiation: 1000 W/m<sup>2</sup>

Authoritative design outdoor air temperature: 25 °C and 23 °C

Authoritative design outdoor air humidity ratio: 10 g/kg and 10 g/kg

### Conclusions

Calculate the current state of building in the work.

- The model is a model of building before test processes of a computer.
- The model is a model of building before test processes of a computer.
- The model is a model of building before test processes of a computer.

### Keywords

Authoritative design temperature: 25 °C

Authoritative design humidity ratio: 10 g/kg

Authoritative design wind speed: 10 m/s

Authoritative design solar radiation: 1000 W/m<sup>2</sup>

Authoritative design outdoor air temperature: 25 °C and 23 °C

Authoritative design outdoor air humidity ratio: 10 g/kg and 10 g/kg

AI Answers →

| Scenario                       | Mass (kg) | Pressure at 40°C (bar) |
|--------------------------------|-----------|------------------------|
| Current tank (ideal)           | 101.7     | -                      |
| Current tank (real gas)        | 95.2      | -                      |
| Max mass at 200 bar (ideal)    | 814.4     | 270                    |
| Max mass at 200 bar (real gas) | 661.9     | 220                    |
| Safe max mass at 40°C, 200 bar | 595.7     | 200                    |

7 VALUES

Z at 200 bar & 40°C  $\approx$  1.28

For 25°C and 25 bar,  $Z \approx 1.07$  for hydrogen.

### Maximum Additional Mass (Ideal and Real Gas)

At 200 bar, 25°C:

- $Z \approx 1.23$  wiley-vch +1

## Reflection

1. My hand-written answers compared to perplexity (my AI of choice) answers were not too far off, but they were also not the same. This was due to several reasons. The first reason was because the AI tool was treated the given pressure as the system pressure whereas I calculated the system pressure or absolute pressure by adding the gauge pressure and the atmospheric pressure. The second reason why our answers were different was because of the  $z$  values. Although they weren't too far off, this still caused a difference in solution.
2. The ideal gas assumption led to significant error because of the  $z$  value. The ideal gas law assumes that  $z = 1$  whereas the real gas law assumes the  $z$  value is anything but the value 1. In this portion where I had to take the percentage error, the  $z$  value caused a significantly lower mass calculation than the mass calculated in the ideal gas law.
3. There are several safety precautions or systems that can be taken to prevent an incident caused by overfilling or overpressure. These precautions can be as simple as regular inspection and maintenance, but they could also be an input of additional operations such as pressure relief valves that release gas if the system pressure exceeds the safety limit. Another operation that can be added to the system could be pressure sensors that monitor system pressure and enable alarms if levels rise above safety protocols.
4. This case has taught me the importance of calculations when it comes to engineering systems. One wrong value or mistake can cause dangerous and hazardous incidents. It is important to not be careless and double check everything before inputting the data into the system. I also learned that there must be considerations of anything that can go wrong. I learned during my time in a lab that it is actually very useful to be paranoid to prevent anything from going wrong. In some cases, incident prevention can be everything.